進捗報告

中島大河

1. 目的

PCB 設計, プログラミング

2. 今週やったこと 1:PCB 設計

Fig. 1 に示すようにセンサー部分の PCB 設計を改善した。3DPCB であれば、筐体設計と同様の操作で主要部品の配置を行うことが出来た。

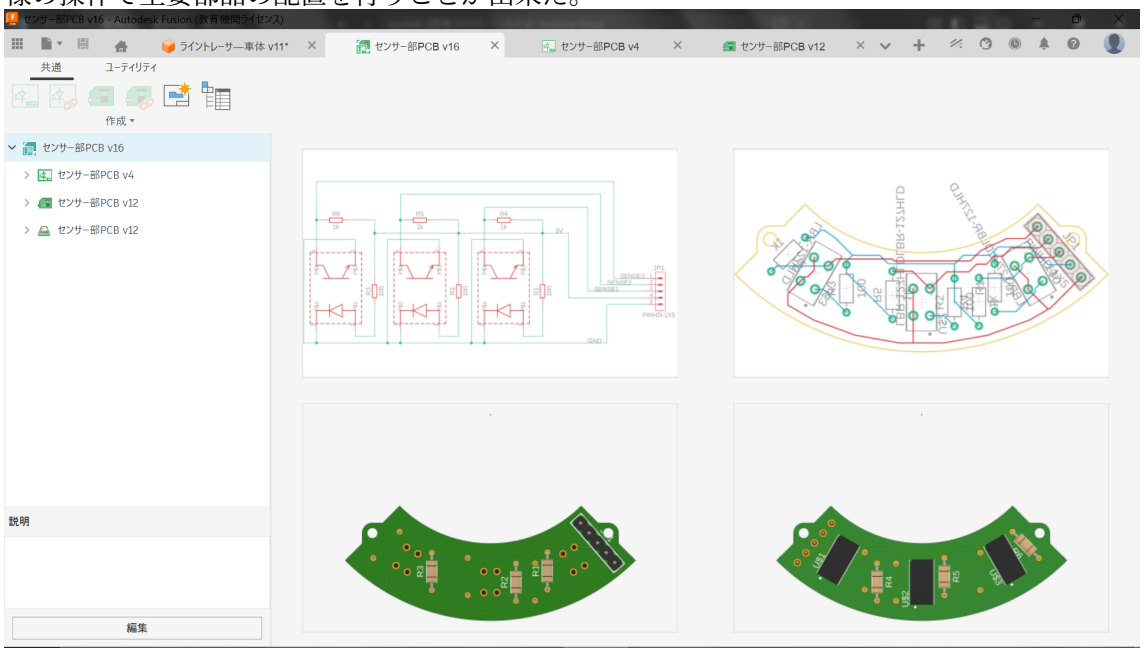

Fig. 1 センサー PCB 設計

原点位置を基準に配置を行うことにより、Fig. 2 の様にセンサーを配置することが出来た。

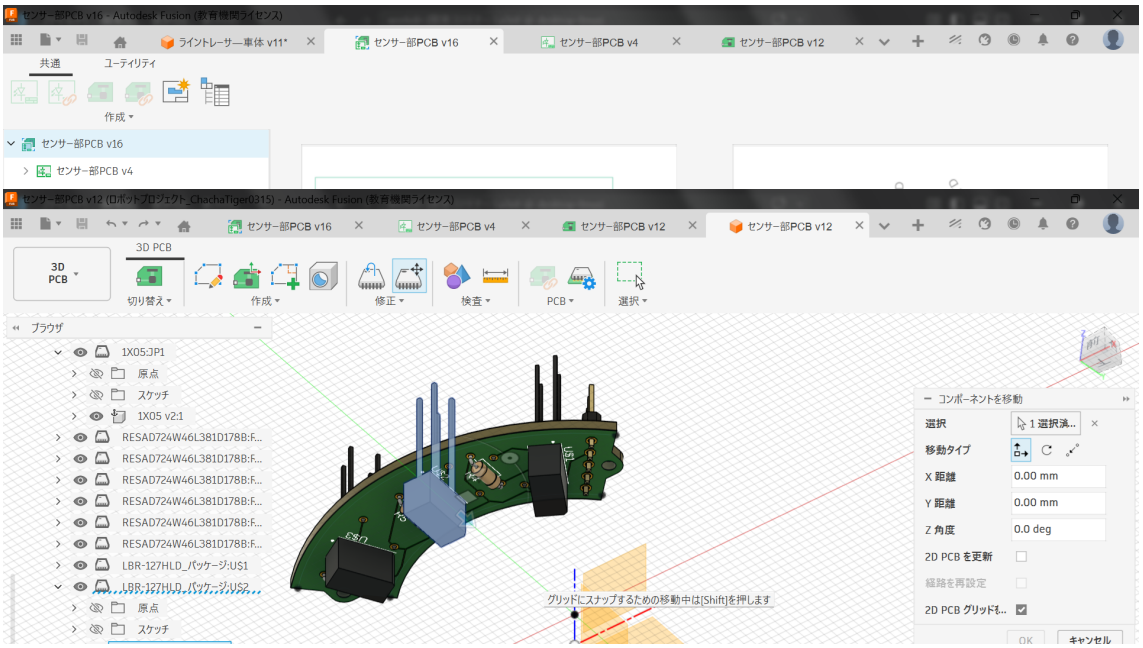

Fig. 2　3DPCB での位置決め

3.　今週やったこと 1: モーター制御クラスの設計

モーター制御専用のクラスを設計した。

```
enum class MotorSide{
  Left,
  Right
};

enum class MotorMode{
  Forward,
  Reverse,
  Coast,
  Brake
};

class Motor{
  public:
    virtual void SetOffset(MotorSide Side,float Value) = 0;
    virtual void SetMultiplier(MotorSide Side,float Value) = 0;
    virtual void SetSpeed(MotorSide Side,MotorMode Mode,float
     ↪  Speed/*0.0~1.0*/) = 0;
    virtual void Drive(float LeftSpeed/*-1.0~1.0*/,float
     ↪  RightSpeed/*-1.0~1.0*/,bool Brake = 0) = 0;
    virtual ~Motor() = default;
};

class MotorDRV8835 final : public Motor{
  public:
```

```
MotorDRV8835(){
    pinMode(LeftMotorPin[0], OUTPUT);
    pinMode(LeftMotorPin[1], OUTPUT);
    pinMode(RightMotorPin[0], OUTPUT);
    pinMode(RightMotorPin[1], OUTPUT);
}

void SetOffset(MotorSide Side,float value) override{
  if(Side == MotorSide::Left)LeftOffset = value;
  else if(Side == MotorSide::Right)RightOffset = value;
}

void SetMultiplier(MotorSide Side,float value) override{
  if(Side == MotorSide::Left)LeftMultiplier = value;
  else if(Side == MotorSide::Right)RightMultiplier = value;
}

void SetSpeed(MotorSide Side,MotorMode Mode,float Speed) override{
  int (&MotorPin)[2] = (Side == MotorSide::Left) ? LeftMotorPin :
  ↪ RightMotorPin;
  float Offset = (Side == MotorSide::Left) ? LeftOffset : RightOffset;
  float Multiplier = (Side == MotorSide::Left) ? LeftMultiplier :
  ↪ RightMultiplier;
  int pwm = constrain((Speed + Offset)*Multiplier*255,0,255);

  switch(Mode){
    case MotorMode::Forward:
      analogWrite(MotorPin[0],pwm);
      analogWrite(MotorPin[1],0);
      break;

    case MotorMode::Reverse:
      analogWrite(MotorPin[0],0);
      analogWrite(MotorPin[1],pwm);
      break;

    case MotorMode::Coast:
      analogWrite(MotorPin[0],0);
      analogWrite(MotorPin[1],0);
      break;

    case MotorMode::Brake:
      analogWrite(MotorPin[0],255);
      analogWrite(MotorPin[1],255);
      break;

    default:
```

```
        analogWrite(MotorPin[0],0);
        analogWrite(MotorPin[1],0);
        break;
      }
    }

    void Drive(float LeftSpeed,float RightSpeed,bool Brake = 0) override{
      if(Brake == true){
        SetSpeed(MotorSide::Left,MotorMode::Brake,0);
        SetSpeed(MotorSide::Right,MotorMode::Brake,0);
      }
      else if(LeftSpeed ==0.0 && RightSpeed == 0.0){
        SetSpeed(MotorSide::Left,MotorMode::Coast,0);
        SetSpeed(MotorSide::Right,MotorMode::Coast,0);
      }
      else{
        SetSpeed(MotorSide::Left,(LeftSpeed > 0.0) ? MotorMode::Forward :
         ↪  MotorMode::Reverse,abs(LeftSpeed));
        SetSpeed(MotorSide::Right,(RightSpeed > 0.0) ? MotorMode::Forward :
         ↪  MotorMode::Reverse,abs(RightSpeed));
      }
    }

  private:
    int LeftMotorPin[2] ={0,1};
    int RightMotorPin[2] = {2,3};
    float LeftOffset = 0.0;
    float RightOffset = 0.0;
    float LeftMultiplier = 1.0;
    float RightMultiplier = 1.0;
};

MotorDRV8835 MotorObj;

void setup() {
  // put your setup code here, to run once:

}

void loop() {
  // put your main code here, to run repeatedly:

}
```

4. まとめ

今回はセンサー部分の PCB 設計とプログラミングを行った.

5.　来週やること

1. 回路設計の続き
2. プログラムの続き